

Isabel Rosa Marie Medlock

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EDUCATION

Yale University

Astronomy PhD Candidate, Thesis Advisor: Prof. Daisuke Nagai
M.Phil. and M.S. in Astronomy

New Haven, CT

Aug 2021 – May 2027

Dec 2024

Princeton University

A.B. in Astrophysical Sciences; Certificates in Applications of Computing
and Russian Language

Princeton, NJ

Sep 2017 – Jun 2021

RESEARCH INTERESTS

My research connects galaxy evolution to precision cosmology by tracing how baryons flow across the circumgalactic medium through the cosmic web, combining simulation-based and analytic models with a suite of observational probes, including fast radio bursts.

PUBLICATIONS

14 publications: 6 first-author (5 refereed, 1 submitted), 6 refereed co-author, 2 non-refereed.

AWARDS & GRANTS

NSF ACCESS Computing Grant (Science-PI): 300k and 312.5k

2025, 2026

Stampede3 node hours (\$62k, \$68.5k)

Yale Dean's Fund for Colloquia and Symposia, for Yale
Cosmology Seminar (\$2,700)

2024 – 2026

Travel Grants: Yale Graduate Student Assembly Conference Travel

2022 – 2025

Fellowship (\$2,400; 2023, 2024, 2025), AAS (\$2,000; 2023, 2024), APS

Division of Astrophysics (\$300; 2023), SACNAS NDiSTEM (\$1,000; 2022)

Dean's Emerging Scholars Research Award, Yale (\$2,000)

2022

TALKS

Invited:

Galaxies Across Cosmic Time in Paso Robles, CA

Oct 2026

Baryons @ Princeton at Princeton University

May 2026

Astronomy Seminar at University of Connecticut

Mar 2026

33rd Texas Symposium: FRB Parallel Session in Tempe, AZ

Dec 2025

ITC Luncheon at Harvard-Smithsonian Center for Astrophysics

Oct 2025

Borthakur Group Meeting at Arizona State University (virtual)

Oct 2025

FRB Journal Club at Max Planck Institute for Radio Astronomy (virtual)

Apr 2025

Astro Lunch at University of Washington

Feb 2025

Selected Contributed:

Summary (2020–present): 26 talks, 4 posters

Filaments in the Multi-Scale Universe at IPMU, Tokyo, Japan

Oct 2026

Dynamic Cosmic Structures in Hangzhou, China	<i>Oct 2026</i>
Cosmic Ecosystems at Perimeter Institute, Waterloo, Canada	<i>Aug 2025</i>
Fast Radio Burst 2025 in Montreal, Canada	<i>Jul 2025</i>
Cosmology and Galaxy Astrophysics with Simulations and ML at CCA	<i>Dec 2024</i>
Fast Radio Burst 2024 in Khao Lak, Phang Nga, Thailand	<i>Nov 2024</i>
Baryons in the Universe 2024 at Kavli IPMU, Kashiwa, Japan	<i>Apr 2024</i>

STUDENT MENTORING

Summary: 1 graduate and 1 undergraduate mentee; 1 co-authored publication

Donglin Wu, Yale Undergraduate Student	<i>Aug 2025 – present</i>
• FRB Host Halo Dispersion Measures in CAMELS-TNG50, paper in preparation.	
Chloe Neufeld, Yale Graduate Student	<i>Sep 2023 – Dec 2024</i>
• Feedback energetics in CAMELS, second author on Medlock et al. 2025, ApJ 980, 61.	

PROFESSIONAL SERVICE

Journal Referee (ApJ, OJA, A&A; 3 Manuscripts)	<i>2026 – present</i>
Yale Cosmology Seminar Co-Organizer	<i>Sep 2023 – present</i>
SACNAS Yale Chapter Co-President, Secretary, and Treasurer	<i>Jan 2023 – present</i>
Astronomy Climate and Diversity Committee Member	<i>Sep 2022 – present</i>
First-Year Astronomy Buddy (FAB) Program Mentor	<i>Aug 2024 – May 2026</i>
Yale Astronomy Siblings Mentor	<i>Sep 2022 – present</i>

OUTREACH

LFOP Spanish Language Night Creator and Organizer	<i>Feb 2023 – present</i>
Leitner Family Observatory and Planetarium (LFOP) Presenter for School Visits	<i>Sep 2022 – present</i>
Science in the News Presenter	<i>2022 – 2023</i>

TEACHING EXPERIENCE

Yale Pathways Instructor for Local High School Students	<i>2025, 2026</i>
Eduexplora Yale Summer Session Instructor	<i>2023</i>
Teaching Fellow, Yale: Galaxies and the Universe; Frontiers and Controversies in Astrophysics; Intro to Relativity and Black Holes; Planets and Stars	<i>2021 – 2023</i>

REFERENCES

Daisuke Nagai (PhD Advisor): Professor of Physics and of Astronomy, Yale University
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Frank van den Bosch (Collaborator): Professor of Astronomy and Physics, Yale University
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Volker Springel (Collaborator): Director, Max Planck Institute for Astrophysics
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Isabel Medlock: Publications

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FIRST AUTHOR

Published and Submitted:

- [6] **Medlock, I.**, and Nagai, D., “Beyond Feedback: Disentangling Baryonic Effects with tSZ x FRB Cross-Correlations”, submitted to *ApJ*
- [5] **Medlock, I.**, Nagai, D., Mandelker, N., Springel, V., van den Bosch, F., Zinger, E., and Chiang, B., “Statistical Properties of Cold Streams in Massive Star-Forming Halos in IllustrisTNG-50”, 2025, *ApJ*, 1000(2), 222
- [4] **Medlock, I.**, Nagai, D., Anglés-Alcázar, D., and Gebhardt, M., “Constraining the Effect of Baryonic Feedback on the Matter Power Spectrum with Fast Radio Bursts”, 2025, *ApJ*, 983(1), 46
- [3] **Medlock, I.**, Neufeld, C., Nagai, D., Anglés-Alcázar, D., Genel, S., Oppenheimer, B., Singh, P., and Villaescusa-Navarro, F., “Quantifying Baryonic Feedback on Warm-Hot Circumgalactic Medium in CAMELS Simulations”, 2025, *ApJ*, 980(1), 61
- [2] **Medlock, I.**, Nagai, D., Singh, P., Oppenheimer, B., Anglés-Alcázar, D., and Villaescusa-Navarro, F., “Probing the Physics of the Circum-galactic Medium using Fast Radio Bursts: Insights from CAMELS”, 2024, *ApJ*, 967(1), 32
- [1] **Medlock, I.**, and Cen, R., “Dispersion Measure Distributions of Fast Radio Bursts Due to the Intergalactic Medium”, 2021, *MNRAS*, 502(3), 3664–3669

REFEREED CO-AUTHOR

Published and Submitted:

- [6] Sims, X., Anglés-Alcázar, D., Oh, B. K., Nagai, D., Mercedes-Feliz, J., **Medlock, I.**, Ni, Y., Lovell, C., and Villaescusa-Navarro, F., “CAMELS Environments: The Impact of Local Neighbours on Galaxy Evolution across the SIMBA, IllustrisTNG, ASTRID, and Swift-EAGLE Simulations”, 2026, *MNRAS*, 551(4), 1529
- [5] Gebhardt, M., Anglés-Alcázar, D., Genel, S., Nagai, D., Oh, B. K., **Medlock, I.**, Mercedes-Feliz, J., Sutherland, S., Lee, M., Sims, X., Lovell, C., Spergel, D., Davé, R., Schaller, M., Schaye, J., and Villaescusa-Navarro, F., “Cosmological Back-Reaction of Baryons on Dark Matter in the CAMELS Simulations”, 2026, *MNRAS*, 547(4), 525
- [4] Oppenheimer, B., Voit, M., Bahé, Y., Battaglia, N., Bregman, J., Burchett, J., Eckert, D., Faerman, Y., Gibson, J., Hummels, C., **Medlock, I.**, Nagai, D., Putman, M., Qu, Z., Sun, M., Werk, J., and Zhang, Y., “The Descriptive Parametric Model I: Gaseous Profiles for Galaxies, Groups, and Clusters”, 2025, *MNRAS*, 543(3), 2649–2669 (*FRB Model lead*)
- [3] Zhang, Z., Nagamine, K., Oku, Y., Lee, K.-G., Fukushima, K., Tomaru, K., Zhang, B., **Medlock, I.**, and Nagai, D., “Probing the Cosmic Baryon Distribution and the Impact of AGN

Feedback with FRBs in CROCODILE Simulation”, 2025, *ApJ*, 993(2), 162

[2] Leung, C., Simha, S., **Medlock, I.**, Nagai, D., Masui, K., Kahinga, L., Lanman, A., Andrew, S., Bandura, K., Curtin, A., Gaensler, B., Gusinskaia, N., Joseph, R., Lazda, M., Mas-Ribas, L., Meyers, B., Nimmo, K., Pearlman, A., Prochaska, J. X., Sammons, M., Shin, K., Smith, K., and Wang, H., “Stellar Mass-Dispersion Measure Correlations Constrain Baryonic Feedback in Fast Radio Burst Host Galaxies”, 2025, *ApJL*, 991(1), L25 (*Simulation analysis lead*)

[1] Lau, E., Nagai, D., Bogdán, Á., **Medlock, I.**, Oppenheimer, B., Battaglia, N., Anglés-Alcázar, D., Genel, S., Ni, Y., and Villaescusa-Navarro, F., “X-raying CAMELS: Constraining Baryonic Feedback in the Circum-Galactic Medium with the CAMELS Simulations and eRASS X-ray Observations”, 2025, *ApJ*, 984(2), 190

NON-REFEREED CO-AUTHOR

[2] Oppenheimer, B., Nagai, D., Lau, E., Singh, P., Butler Contreras, A., Gluck, N., Dorigo Jones, J., **Medlock, I.**, and Villaescusa-Navarro, F., “A Multi-Wavelength, Multi-Model Exploration of How Feedback Disrupts Gaseous Atmospheres”, 2022, *Bulletin of the AAS*, 54(1)

[1] Singh, P., Nagai, D., Oppenheimer, B., Lau, E., Gluck, N., and **Medlock, I.**, “Galactic Gaseous Halos: Mini-Clusters Disrupted by Feedback”, 2022, *Galactic Atmospheres*